

USB THUMB DRIVES

By virtue of their extremely small sizes and low weight, USB thumb drives are easily the most portable storage solutions available today. A discernible benefit with these devices is the absence of any moving parts whatsoever, which means almost no maintenance issues as compared to other storage solutions. Though older thumb drives suffered from low storage volumes, the advances in memory technology are changing this scenario for the better.

Features

We received 16 models, with data capacities varying from the abysmally low Freecom 32 MB drive, to the absolutely monstrous 1 GB drive from Princeton. So while analysing the features of each drive, we did not consider the storage capacity, but paid more attention to the physical characteristics and build-quality in relation to its effects on the mobility of the drive.

As far as physical dimensions are concerned, all the drives were quite small and had relatively low weight. The lightest drives we received were the 64 MB and 128 MB models from Princeton, which weighed 8 gms. The heavier models included the 512 MB drive from Transcend, which weighed around 24 gms, as well as the solitary 20-gm Freecom FM-1. The extra weight of the FM-1 was due to its sturdy and durable design.

The EasyDisk 128 MB was the clear winner; primarily due to its decent software bundle and its good scores in our tests



Most other models sacrificed durability for weight, using poorer material to get better visual appearance and lower weight. Some drives did manage to attain a decent balance between good design and appearance. The Nexdisk and Transcend drives had exceptional designs and an attractive silver finish.

All the drives had a single LED to indicate when they were powered and transferring data.

One major letdown with all the drives was that they used the slower USB 1.1 interface—all except the 1 GB beauty from Princeton, which uses USB 2.0. The advantage of USB 2.0 is obvious in the performance tests, as the Princeton 1 GB simply obliterated the competition. It's sad to see that manufacturers are slow to incorporate the USB 2.0 technology that has been around for quite a while now. The fact that USB 2.0 data transfer speeds are as much as 40 times higher than USB 1.1, should be enough reason to make the switch.

Another important aspect we considered was the presence of inbuilt security functions. As most of these devices are tiny, they can be easily misplaced or stolen—very troublesome if the drive contains sensitive documents and files. Another need for security could arise if the drive is used by two or more people: you lend the drive to your friend to transfer documents to his or her computer, but you don't want them modifying or deleting anything, accidentally or otherwise. Almost all the drives we tested had a simple and accessible write-protection switch on them. The Touchmate TM-USB256 had a small switch inside a groove



How We Tested

Due to the varying type of the portable storage devices we received, we had to take different approaches during our test process. Our test configuration, however, remained the same with a P4 3.06 GHz processor with Hyper-Threading enabled on an i845 chipset-based ABit BG7E motherboard and 256 MB RAM. A Seagate 40 GB hard disk was utilised as the primary system storage device, and was clean-formatted before loading Windows XP. All the latest drivers for the motherboard and USB ports were loaded. To maintain a standard environment, we installed only the required benchmarking utilities and the files that we used in our tests.

Test Methodology

All the storage devices we tested were evaluated on the basis of their scores in three primary factors, namely features, performance and value for money. To help in our calculations, we assigned a relevant amount of percentage points to these parameters. These parameters included:

Features: Features made up 15 per cent of the total score. We looked at the various inbuilt characteristics and qualities of each gadget. Considering that every product was marketed as a portable device, we laid particular emphasis on the drive's weight and dimensions. We also observed the build-quality, with primary importance given to the sturdiness of the product. We gave extra points for any security features such as a write-protection switch, and also for useful software in the package.

Performance: To gauge the performance—which made up 85 per cent of the score—we conducted two types of tests—synthetic and real-world. For the synthetic tests, we used the SiSoft Sandra Professional 2003 benchmarking suite to assess theoretical read and write speeds, as well as the average access time. We also tested with the more specific HDTach 2.7 benchmark and noted the average CPU utilisation of each drive.

In the real-world tests, we transferred a set of assorted files of varying sizes totalling up to 1 GB, back and forth, from the system hard disk to the external hard disk, and noted the time taken during each transfer. Similarly, for noting the sequential speeds, we transferred a large image of 1 GB between the system and the external drive.

For the thumb drives, the same test process was followed, with only the size of the test files reduced to 20 MB. Due to the varying storage capacities of each model, we also carried out an additional test, where the thumb drive was filled to its maximum size with both sequential as well as assorted data. We observed the time taken for each transfer, and then calculated the actual data transfer rate in terms of megabytes transferred per second.

Value for money: We noted the actual capacity, and not the rated capacity, of each drive after formatting, and divided it by its price. This gave us a cost index of price per MB value, indicating the cost-effectiveness of each product. Finally, to obtain the VFM quotient, we divided the sum total of our performance and features by the cost index we just calculated.